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BreastDCEDL: A standardized deep learning-ready breast DCE-MRI dataset of 2,070 patients

Naomi Fridman[1]✉, Bubby Solway[2], Tomer Fridman[2], Itamar Barnea[2] & Anat Goldstein[1]

Dynamic Contrast-Enhanced Magnetic Resonance Imaging (DCE-MRI) is essential for monitoring breast cancer treatment response, yet deep learning progress is limited by the lack of standardized, multi-center datasets. We present BreastDCEDL, a deep learning-ready dataset comprising pretreatment 3D DCE-MRI scans from 2,070 patients across three cohorts: I-SPY1 (n = 172), I-SPY2 (n = 982), and Duke (n = 916). Raw DICOM files were converted to standardized 3D NIfTI format, preserving signal integrity and spatial resolution. The dataset includes unified tumor annotations—binary masks for I-SPY cohorts and bounding boxes for Duke—and harmonized clinical metadata. Pathologic complete response (pCR) data are available for 1,452 patients (100% of I-SPY1/I-SPY2, 32.5% of Duke). Clinical biomarkers include HR status (64.1% positive) and HER2 status (22.2% positive) for over 99% of patients. Predefined train-validation-test splits (1,532/268/270) maintain balanced distributions. BreastDCEDL fills a critical gap in public imaging resources, supporting development of advanced models for pCR prediction and other breast cancer analyses. The dataset is designed for broad reuse and is accompanied by processing tools to support community-driven AI research in oncology.

Background & Summary

Breast cancer is a leading cause of cancer death worldwide. Early detection and accurate treatment monitoring are crucial to better patient outcomes. An important clinical decision after diagnosis is the choice between immediate surgery or neoadjuvant chemotherapy (NAC) before surgery. NAC aims to reduce tumor size, enhancing the potential for surgical success. The optimal outcome is the complete pathologic response (pCR), defined as the absence of residual invasive disease following therapy, confirmed by postsurgical biopsy. Accurate prediction of pCR is critical because it strongly correlates with improved long-term survival and can guide personalized treatment decisions. Pre-treatment pCR probability estimates could support clinical decision-making in several ways: risk stratification for therapy intensification or de-escalation, clinical trial enrichment by identifying patients most likely to benefit from novel agents, and early identification of likely non-responders who may benefit from alternative treatment strategies. Prior work suggests that MRI acquired during or after NAC can outperform pre-treatment imaging for pCR prediction, and that combining imaging with clinicopathologic variables further improves performance[1]. BreastDCEDL includes harmonized clinical metadata to support multimodal analyses; conversion of mid-treatment I-SPY timepoints is planned for future releases. Dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) is a principal imaging modality for the evaluation of breast cancer. The technique involves acquiring sequential 3D scans before and after the administration of contrast agents, typically analyzed at three time points: baseline before contrast, early post-contrast (maximum enhancement) and late post-contrast (washout dynamics), as shown in Fig. 1. These acquisitions enable visualization of abnormal tumor vascularity—characterized by increased vessel density, permeability, and disorganized structure—that distinguishes malignant tissues from normal breast parenchyma[2–4]. Deep learning applied to breast DCE-MRI analysis has shown promise, but faces limitations. Multiple systematic reviews[5–8] have identified critical barriers, including insufficient access to large-scale datasets, inconsistent evaluation protocols, and inadequate external validation. These works span from technical deep learning reviews (Houssein[6], Wei[7]) to modality-specific MRI analyses (Gullo[5], LoGullo[8]), collectively highlighting the need for standardized datasets and reproducible benchmarks to advance clinical translation. Although convolutional neural networks have shown promising results through multiphase fusion[9], multimodal integration[10,11], and multi-institutional

[1]Department of Industrial Engineering, Ariel University, Ariel, Israel. [2]NF Algorithms & AI, Tel Aviv, Israel. ✉e-mail: naominoe.fridman@msmail.ariel.ac.il

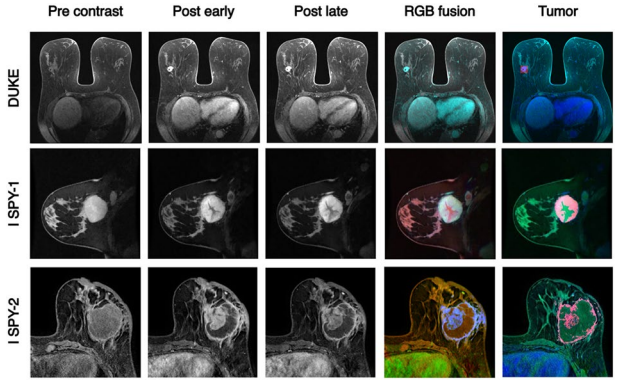


Fig. 1 Dynamic Contrast-Enhanced MRI illustrates tumor segmentation across datasets. Representative DCE-MRI from the Duke, I-SPY1, and I-SPY2 datasets show pre-contrast, early post-contrast, and late post-contrast acquisitions. Columns display individual phases (left), a merged RGB composite (center), and tumor segmentation (right). Tumor localization in the Duke dataset is highlighted by a bounding box.

validation[12], most investigations remain constrained by limited sample sizes and non-standardized data formats. A recent survey of publicly available MRI datasets for AI research[13] identified 11 breast MRI collections, of which only the I-SPY1, I-SPY2, and Duke cohorts provide substantial patient numbers with NAC treatment data and pCR outcomes. Other breast MRI datasets are either substantially smaller (e.g., QIN-Breast: 67 patients, Breast-MRI-NACT-Pilot: 64 patients) or focus on diagnosis rather than treatment response prediction. To address these challenges, we present BreastDCEDL, a comprehensive deep learning-ready dataset for breast DCE-MRI. By integrating data from I-SPY2[14–16] (985 patients, 3 excluded for lacking three timepoints; n = 982), Duke[17,18] (922 patients, 6 excluded for lacking three timepoints or bounding boxes; n = 916), and I-SPY1[19,20] (221 patients, of which 172 had at least three scans), we have created a standardized resource comprising 2,070 patients. The dataset transforms complex digital imaging and communications in medicine (DICOM) structures into accessible 3D arrays with preserved signal integrity, unified tumor annotations, and harmonized clinical metadata. Through BreastDCEDL, our objective is to accelerate progress in AI-assisted breast cancer care by providing the research community with the resources needed to develop, validate, and compare deep learning models in a standardized environment.

Methods

BreastDCEDL Dataset Composition. BreastDCEDL integrates three datasets from The Cancer Imaging Archive (TCIA)[21]: I-SPY2 (Investigation of Serial Studies to Predict Your Therapeutic Response with Imaging And moLecular Analysis 2)[14–16], I-SPY1 (QuantumLeap Healthcare Collaborative)[19,20], and Duke[17,18]. I-SPY2 contains 985 patients (3 excluded for lacking three timepoints), I-SPY1 has 172 patients with at least three scans, and Duke contains 916 patients with complete DCE data. This integration resulted in a cohort of 2,070 patients (Table 1).

Table 2 summarizes the institutional and scanner distribution across datasets.

DICOM to NIfTI Conversion Pipeline. The transformation from clinical DICOM to machine learning-ready NIfTI required addressing significant heterogeneity across 22+ imaging centers. The original data comprised 8.5 million DICOM slices (4.6TB) organized in hierarchical folders with mixed MRI acquisitions and vendor-specific tags, which we converted to 11,717 standardized NIfTI volumes (183GB) with clear naming convention: `<id>_<ds>_acq<N>.nii.gz`.

Temporal Reconstruction. DCE-MRI temporal information was encoded differently across vendors. We developed a hierarchical parsing strategy with three levels: (1) Primary identification using clinical trial timepoint tags (0012,0050) for T0 verification; (2) Contrast timing extraction from private tags (0117,1030-1035) including total phases (0117,1030), injection time (0117,1033), and SER timing indices (0117,1035); (3) Fallback parsing using SeriesDescription and ProtocolName when standard tags were absent.

For I-SPY2, tag (0117,1033) provided injection timing for precise alignment of pre-contrast (index 0), peak enhancement (index 2), and washout phases (indices 5–7). Duke data used "Pre", "Post1", "Post2" naming conventions in SeriesDescription fields.

Spatial Organization. DICOM slices arrived non-sequentially in 68% of series. Our pipeline: (1) extracted slice location from ImagePositionPatient (0020,0032); (2) computed slice normal vectors from ImageOrientationPatient (0020,0037); (3) sorted slices along anatomical axis using dot product projection; (4) validated against I-SPY analysis masks as spatial ground truth.

This automated pipeline eliminated 20-30 hours of manual preprocessing per researcher compared to raw TCIA data processing.

Section	Characteristic	Duke (n = 916)	I-SPY1 (n = 172)	I-SPY2 (n = 982)	Overall (n = 2070)
Demographics					
	Age (years), mean	52.4	48.0	48.8	50.3
	Race: Black (%)	200 (21.8)	28 (16.3)	117 (11.9)	345 (16.7)
	Race: White (%)	648 (70.7)	123 (71.5)	785 (79.9)	1556 (75.2)
HR Status					
	HR+ (%)	696 (76.0)	95 (55.9)	536 (54.6)	1327 (64.2)
	HR− (%)	220 (24.0)	75 (44.1)	446 (45.4)	741 (35.8)
HER2 Status					
	HER2+ (%)	163 (17.8)	53 (30.8)	242 (24.6)	458 (22.1)
	HER2− (%)	753 (82.2)	119 (69.2)	740 (75.4)	1612 (77.9)
Receptor Subtype					
	HR+/HER2+ (%)	104 (11.4)	10 (5.8)	155 (15.8)	269 (13.0)
	HR+/HER2− (%)	592 (64.6)	66 (38.4)	381 (38.8)	1039 (50.2)
	HR−/HER2+ (%)	59 (6.4)	43 (25.0)	87 (8.9)	189 (9.1)
	Triple Negative (%)	161 (17.6)	43 (25.0)	359 (36.6)	563 (27.2)
pCR					
	pCR+ (%)	63 (21.1)	49 (28.5)	316 (32.2)	428 (29.5)
	pCR− (%)	235 (78.9)	123 (71.5)	666 (67.8)	1024 (70.5)

Table 1. Patient demographic characteristics and biomarker statuses in the BreastDCEDL database. Age is reported as mean. Values in parentheses are percentages. pCR percentages are based on patients with available data (Duke: 298/916, I-SPY1: 172/172, I-SPY2: 982/982).

Dataset	Patients	Institutions	GE	Siemens	Philips
I-SPY1	222	10	145 (65.3%)	64 (28.8%)	13 (5.9%)
I-SPY2	985	>22	613 (62.2%)	253 (25.7%)	119 (12.1%)
Duke	922	1	628 (68.1%)	294 (31.9%)	–

Table 2. Scanner and institutional distribution across source datasets.

Tumor Annotation Processing. *I-SPY Trials Segmentation.* I-SPY trials contained complex processed data including ROI delineations, SER images, enhancement maps, and analysis masks encoding segmentation through byte operations. Segmentation steps were encoded as bit flags: manual VOI selection (1), thresholding (32), with combinations like code 33 (32+1) indicating voxels within VOI above threshold. We decoded these bitmaps into binary 3D masks (1 = malignant, 0 = background), yielding 172 masks for I-SPY1 and 982 for I-SPY2.

Duke Dataset Annotations. Duke sequences included non-fat saturated and fat-saturated scans at 1–2 minute intervals. Expert radiologists annotated bounding boxes for the largest tumor (among up to 9 per patient) based on pre-contrast DCE-MRI, first post-contrast, and their subtraction. We transformed these coordinates to align with standardized DCE-MRI orientation for 916 patients. Temporal resolution is approximately 60 seconds for Duke and 120 seconds for I-SPY trials; to harmonize early post-contrast timing across cohorts, we selected Duke's second post-contrast acquisition (index 2) to approximate I-SPY early post-contrast timing.

Data Type Preservation and Standardization. Original DICOM used 16-bit integer or unsigned integer representation. We preserved full dynamic range by converting to 64-bit floating point format, maintaining signal integrity for quantitative analysis (Table 3).

Clinicopathologic Data Harmonization. We documented biomarker definition variations across datasets. HR status determination varied: I-SPY2 used greater than or equal to 1% staining, Duke used Allred score greater than 3, and I-SPY1 required greater than or equal to 10% staining. HER2 status assessment differed with Duke using FISH ratio greater than 2.2 versus I-SPY trials using FISH ratio greater than or equal to 2.0.

Original classifications were preserved in metadata files, enabling consistent threshold application. Tumor volumes were calculated by multiplying voxel counts by physical dimensions (I-SPY) or pre-categorized into size classes (Duke).

Benchmark Establishment. Fixed train-validation-test partitions ensure reproducible benchmarking across all 2,070 patients. Training set contains 1,532 patients total (1,099 with pCR data, 29.3% pCR rate). Validation set contains 268 patients total (177 with pCR data, 29.9% pCR rate). Test set contains 270 patients total (176 with pCR data, 30.1% pCR rate).

Aspect	Original TCIA	BreastDCEDL
Format	8.5M DICOM slices (4.6TB)	11,717 NIfTI volumes (183GB)
Organization	DICOM files of slices and 3D data,	3D NIfTI file per acquisition
	mixed modalities and acquisitions	with standardized naming:
	with vendor-specific tags	`<id>_<ds>_acq<N>.nii.gz`
Temporal	Vendor and clinic-specific	Standardized temporal indices
alignment	custom tags per slice	
Segmentations	Bitmaps encoding	Binary 3D masks (1,154) +
	segmentation procedures	Bounding boxes (916)
Metadata	Embedded in 8.5M DICOM headers	Single unified CSV file
Load time	Approximately 5–10 minutes	Less than 5 seconds
per patient		

Table 3. Comparison of original TCIA format versus BreastDCEDL standardized format, detailing differences in data representation and metadata fields.

The 1,452 patients with available pCR data are balanced across splits, while the remaining 618 patients (primarily from Duke without NAC) are included for unsupervised learning and segmentation tasks.

Computational Requirements. Data standardization pipeline: 48 hours on workstation with 64 GB RAM.

Data Records

We developed BreastDCEDL, a comprehensive and standardized DCE-MRI dataset comprising 2,070 patients from three major clinical cohorts: Duke[17,18] (n = 916), I-SPY1[19,20] (n = 172) and I-SPY2[14–16] (n = 982). The processed dataset is deposited on Zenodo[22]. The dataset includes:

- **I-SPY2**: 7,040 DCE-MRI volumes (38.2 GB), 982 segmentation masks (59.1 MB)
- **I-SPY1**: 585 DCE-MRI volumes (3.19 GB), 172 segmentation masks (32.1 MB)
- **Duke**: 4,105 DCE-MRI volumes (142 GB), 916 bounding box annotations in CSV format

All data have been converted to standardized 3D NIfTI format with preserved DICOM intensity values, spatial resolution, and imaging parameters. Each patient includes 3–12 temporal 3D volumes with corresponding annotations and harmonized clinical metadata. The data organization structure and representative examples from each cohort are presented in Fig. 2.

The dataset includes predefined train-validation-test partitions encompassing all 2,070 patients:

- **Training**: 1,532 patients (1,099 with pCR data: 29.3% pCR rate)
- **Validation**: 268 patients (177 with pCR data: 29.9% pCR rate)
- **Test**: 270 patients (176 with pCR data: 30.1% pCR rate)

Among the subset of 1,452 patients with available pCR outcomes (primarily from I-SPY trials), the response rates are balanced across splits. The remaining 618 patients (mainly from Duke cohort without neoadjuvant treatment) are included in the splits for unsupervised learning, segmentation tasks, and radiomic feature development.

Table 4 summarizes the harmonized metadata fields. Spatial parameters (voxel dimensions, image size) are included to facilitate spatial harmonization. Acquisition-specific parameters such as scanner vendor, field strength, TR, and TE can be extracted from the original TCIA DICOM files using utilities provided in our GitHub repository.

Technical Validation

Data Transformation Validation. We successfully transformed 8,543,482 individual DICOM slices from TCIA into 11,717 analysis-ready 3D NIfTI volumes:

- 100% successful reconstruction of 3D volumes from 2D DICOM slices
- Preservation verified for original 16-bit intensity values in 64-bit NIfTI format
- Spatial ordering validated through anatomical landmark checking
- Temporal sequence alignment confirmed across heterogeneous acquisition protocols

Data Completeness. Table 5 demonstrates the comprehensive nature of the BreastDCEDL dataset across all three cohorts. All 2,070 patients have complete DCE-MRI acquisitions successfully converted to NIfTI format. Tumor annotations are available for every patient, with I-SPY cohorts providing 1,154 volumetric segmentation masks and Duke providing 916 bounding box annotations. Clinical biomarker data show excellent completeness, with HR status available for 99.9% of patients (2,068/2,070) and HER2 status for 99.8% (2,065/2,070). The primary limitation concerns pathologic complete response data, which is available for all I-SPY patients (n = 1,154) but only 32.5% of Duke patients (298/916) due to differences in treatment protocols, resulting in 70.1% overall

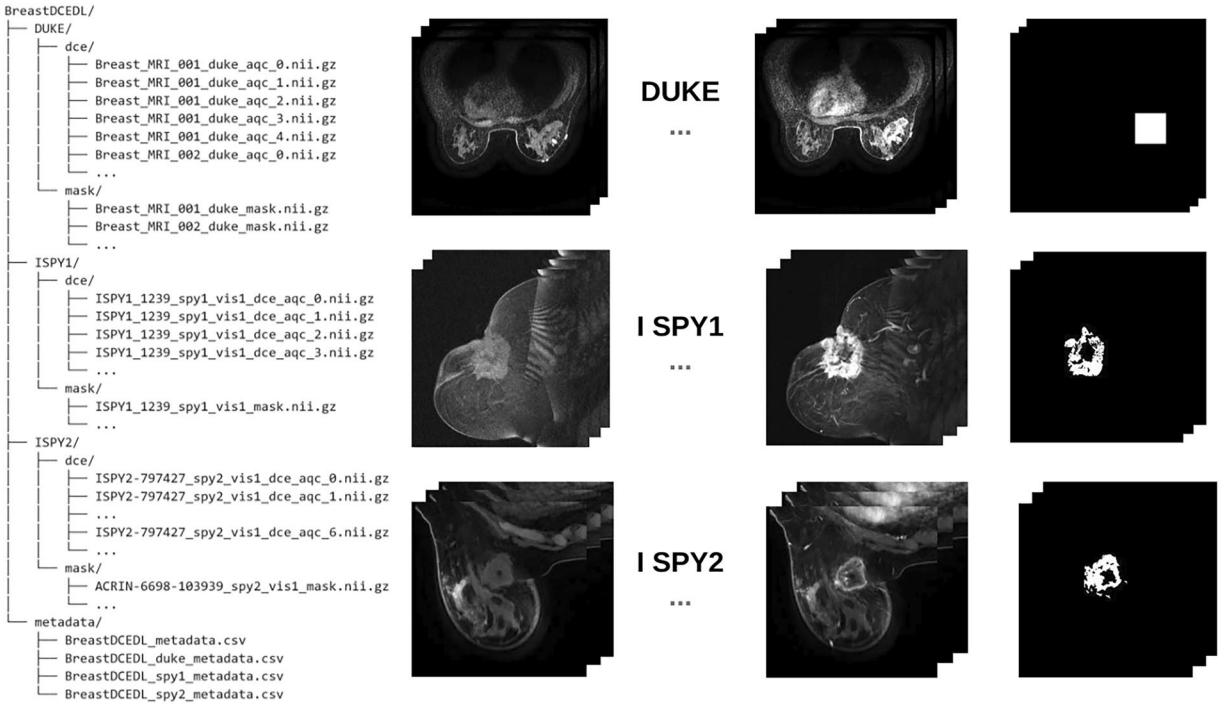


Fig. 2 BreastDCEDL structure and imaging examples. Standardized 3D NIfTI format with clinical annotations facilitates organized data integration. Representative DCE-MRI examples from contributing cohorts illustrate diverse imaging characteristics across institutions.

pCR data availability (1,452/2,070). This distribution reflects the Duke cohort's inclusion of both neoadjuvant and primary surgery cases, while I-SPY trials exclusively enrolled neoadjuvant chemotherapy patients.

Annotation Validation. Tumor segmentation masks were validated for spatial alignment with DCE-MRI volumes. All 1,154 binary masks correctly overlapped with tumor regions. Tumor volumes ranged from 0.5-187.3 cm^3 (median 8.7 cm^3), consistent with clinical expectations for breast lesions.

Data Access and Implementation. The dataset's standardized NIfTI format ensures compatibility with established neuroimaging workflows. Each patient's temporal DCE series follows a consistent naming convention (`<id>_<ds>_acq<N>.nii.gz`) with corresponding tumor segmentations (`<id>_mask.nii.gz`), enabling automated batch processing without manual file matching. Standard libraries including nibabel, SimpleITK, and ITK-SNAP can directly load these volumes without format conversion.

Comprehensive implementation resources are maintained at https://github.com/naomifridman/BreastDCEDL, providing Python utilities for data loading and metadata parsing, temporal alignment verification scripts, Jupyter notebooks demonstrating analysis workflows and data visualization.

Dataset Characteristics and Limitations. BreastDCEDL preserves the inherent heterogeneity of multi-institutional clinical data, which enhances generalizability but requires careful consideration during analysis. The dataset encompasses significant variation in imaging protocols, with temporal resolution ranging from 60 seconds (Duke) to 120 seconds (I-SPY trials), potentially affecting kinetic curve analysis. Biomarker definitions also vary due to institutional standards: HR positivity thresholds range from $\geq$1% (I-SPY2) to $\geq$10% (I-SPY1), while HER2 positivity criteria differ between FISH ratio $>$2.2 (Duke) and $\geq$2.0 (I-SPY trials). No intensity harmonization or spatial normalization was applied to the converted data; original acquisition characteristics are preserved to allow researchers flexibility in applying harmonization methods appropriate to their specific analyses.

Annotation scope represents an important consideration for segmentation tasks. Only primary tumors are annotated despite multifocal disease being present in numerous cases, with up to nine distinct lesions observed in some Duke patients. This limitation means segmentation models trained exclusively on BreastDCEDL may require additional training data to detect secondary lesions reliably. Furthermore, Duke annotations consist of bounding box coordinates rather than pixel-level masks, necessitating adapted training strategies or conversion procedures for certain applications.

The availability of clinical endpoints varies across the dataset. While all I-SPY trial participants have pCR outcomes due to trial design, only 32.5% of Duke patients have this information, as many underwent primary surgery without neoadjuvant therapy. This distribution provides 1,452 patients with pCR labels suitable for supervised learning, while the remaining 618 patients remain valuable for unsupervised learning, segmentation tasks, and radiomic feature development.

Category	Field	Description
Identifiers		
	pid	Unique patient identifier
	dataset	Source cohort (spy1, spy2, duke)
Spatial Parameters		
	n_xy, n_z	Image dimensions (pixels, slices)
	xy_spacing, slice_thick	Pixel spacing, slice thickness (mm)
Temporal Indices		
	n_times	Number of timepoints
	pre, post_early, post_late	Acquisition indices
Tumor Localization		
	mask_start, mask_end	Tumor slice range (z)
	sraw, eraw, scol, ecol	Bounding box coordinates
	tum_vol	Tumor volume (cm^3)
Demographics		
	age	Patient age (years)
	menopause	Menopausal status
	race_white, race_black	Race indicators
Clinical Biomarkers		
	HR	Hormone receptor status (0/1/NaN)
	HER2	HER2 status (0/1/NaN)
	pCR	Pathologic complete response (0/1/NaN)
Data Splits		
	test	Split assignment (0=train, 1=test, 2=val)

Table 4. Metadata fields included in the harmonized CSV file. Spatial parameters enable image harmonization; clinical variables support predictive modeling. Acquisition-specific parameters (scanner vendor, field strength, TR, TE) can be extracted from original TCIA DICOM files using provided code utilities.

Metric	I-SPY2	I-SPY1	Duke	Total
Patients	982	172	916	2,070
Complete DCE	982 (100%)	172 (100%)	916 (100%)	2,070 (100%)
Annotations	982 masks	172 masks	916 BBox	2,070 total
pCR Status	982 (100%)	172 (100%)	298 (32.5%)	1,452 (70.1%)
HR Status	982 (100%)	170 (98.8%)	916 (100%)	2,068 (99.9%)
HER2 Status	982 (100%)	167 (97.1%)	916 (100%)	2,065 (99.8%)

Table 5. Data completeness across datasets. Annotations comprise 1,154 segmentation masks from I-SPY cohorts and 916 bounding boxes from Duke.

Recommended Research Applications. BreastDCEDL's standardized format and comprehensive annotations support diverse research directions in computational breast imaging. The predefined train/validation/test splits with balanced pCR rates (29.3%, 29.9%, and 30.1% respectively) ensure reproducible benchmarking across studies. The dataset is particularly well-suited for developing and validating treatment response prediction models, advancing tumor segmentation algorithms across different annotation types, extracting and validating radiomic biomarkers, studying multi-institutional model generalization, and characterizing temporal enhancement patterns across molecular subtypes.

The inclusion of patients without pCR outcomes enables semi-supervised learning approaches that leverage the full dataset, while the preservation of original intensity values supports quantitative perfusion analysis and pharmacokinetic modeling. Researchers can apply consistent biomarker thresholds across cohorts using the preserved original classifications, enabling both harmonized analyses and studies of inter-institutional variability.

Ethics statement. This study used publicly available deidentified datasets from TCIA[21], including I-SPY1[19,20], I-SPY2[14–16], and Duke Breast Cancer MRI[17,18] datasets. All original data collection for these datasets was conducted with appropriate institutional review board (IRB) approvals at the respective participating institutions, with informed consent obtained from all participants as described in the original publications. No additional ethical approval was required for this secondary analysis of publicly available anonymized data. All data handling and analysis procedures were in accordance with the relevant data protection regulations and institutional guidelines of Ariel University.

Data availability

The BreastDCEDL dataset is available on Zenodo[22]. The repository contains standardized NIfTI volumes for 2,070 patients, including:

- **I-SPY1**: Complete dataset with all DCE-MRI acquisitions and segmentation masks for 172 patients
- **I-SPY2**: Subset containing 3 timepoints (pre-contrast, early post-contrast, late post-contrast) per patient, cropped to 256 × 256 × Z containing tumor regions
- **Duke**: Subset with 3 DCE-MRI timepoints per patient, cropped to 256 × 256 × Z tumor-centered volumes for 916 patients. Harmonized clinical metadata and predefined train-validation-test splits are included

Harmonized clinical metadata and predefined train-validation-test splits are included. The complete I-SPY2 pretreatment NIfTI conversion with full-resolution volumes for all 982 patients is also available on TCIA[23].

Code availability

The code for data processing is available on GitHub (https://github.com/naomifridman/BreastDCEDL) and archived on Zenodo[24]. The repository includes data loading utilities, usage examples, Jupyter notebooks demonstrating data access and visualization, and documentation for reproducing the DICOM-to-NIfTI conversion pipeline.

Received: 21 October 2025; Accepted: 8 January 2026;
Published online: 15 January 2026

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Acknowledgements
This research was supported by the Ministry of Innovation, Science & Technology, Israel, under the National Track Research Program (Grant No. 0005463). We gratefully acknowledge The Cancer Imaging Archive (TCIA) for providing access to the I-SPY1, I-SPY2, and Duke Breast Cancer MRI datasets that formed the foundation of this work. The authors thank the investigators and institutions that contributed to the original data collection and made these valuable datasets publicly available for research purposes.

Author contributions
N.F. led the project conception, methodology design, data standardization, and manuscript preparation. B.S., T.F., and I.B. collaboratively developed the data filtering protocols, processing pipelines, GitHub repository infrastructure, and validation procedures. A.G. supervised the project. All authors participated in manuscript revision and approved the submitted version.

Competing interests
The authors declare no competing interests.

Additional information
Correspondence and requests for materials should be addressed to N.F.

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